Spectral Heart Rate Estimation using AS7265x Sensor

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Checkpoint 3

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Our project

Goal

To Investigate and develop a multi-spectral optical sensing heart rate sensor using AS7265x.

Idea

- Measure reflected light from skin at multiple wavelengths
- Observe periodic changes caused by blood flow
- Estimate heart rate from periodic variations in reflected spectral signals.

What changed since Checkpoint 2

Previously:

- Weak/ no useful signal

Discovered Issue:

- Light source was not activated/correctly used.

Current Experiments

Now: testing with illumination enabled.

- Testing selected channels D, F, R.
- Current best setting: Integration cycle 2, gain 64, around 7 Hz max
- Integeration Cycle 1: too weak/no visible signal.
- Slower settings: more noise / lower temporal resolution.

Parameter	**Effect**	**Trade-off**
Integration Cycles	Controls how long the sensor collects light	↑ cycles → stronger signal but ↓ **sampling** rate
Gain	Amplifies the measured signal	↑ gain → stronger signal but ↑ noise

Main Technical Challenge

- AS7265x is designed for spectral accuracy, not high-speed PPG.
- Heart-rate detection needs enough samples over time.
- Current focus: optimize sampling rate vs signal quality.

Using all channels reduces sampling rate significantly. We selected a subset (D, F, R) to balance temporal resolution and spectral information.

```
sensor.takeMeasurements(); //  13 Hz Sampling rate just func Call
```

Current Pipeline

ESP32 + AS7265x → D/F/R channels → Python BLE logger → dashboard → HR ground truth → CSV logging.

Spectral Signals (D, F, R)
↓
Store in Buffer (time window)
↓
Remove Trend (DC component)
↓
Bandpass Filter (0.7 – 2 Hz)
↓
FFT (Frequency Analysis)
↓
Find Dominant Frequency
↓
Convert to BPM (× 60)
↓
Combine Channels Estimates (Median)
↓
Prototype HR

Sampling rate: ~7 Hz

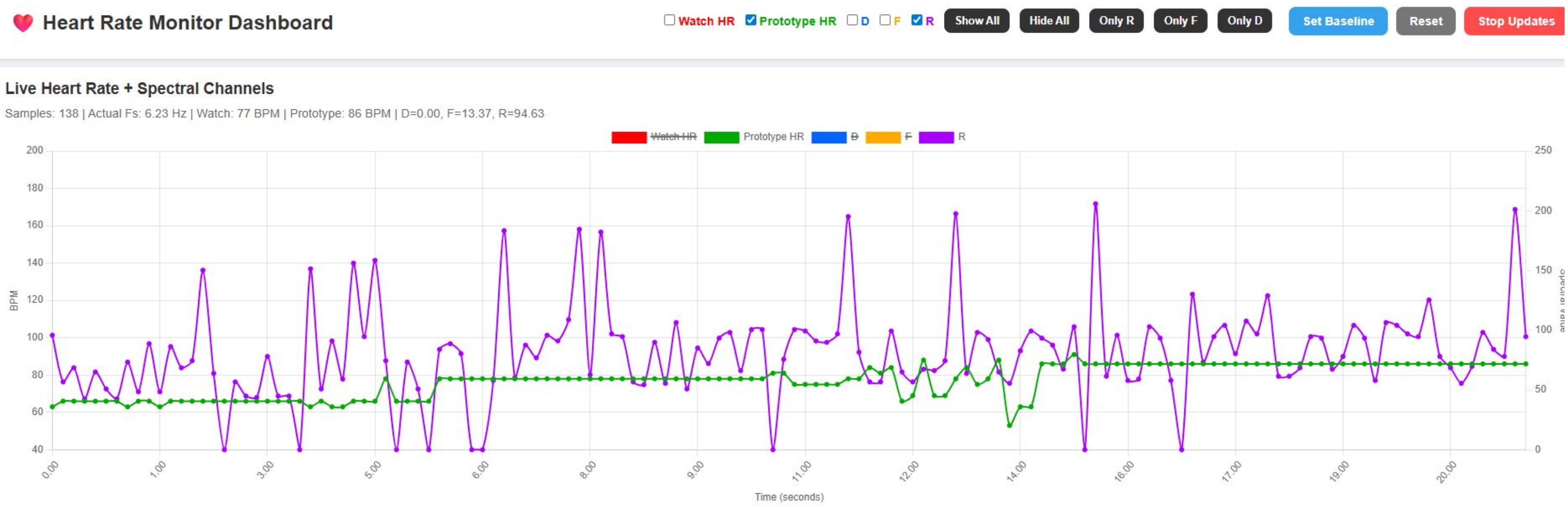
FFT is used because the signal is noisy and frequency-based estimation is more robust than peak detection.

Readings of the ESP32

```
yze_dual_ble.py
2026-04-29T03:03:26] Watch HR:  81 BPM | Prototype HR:  99 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:26] Watch HR:  81 BPM | Prototype HR:  99 BPM | D=0.00, F=0.75, R=4.73
2026-04-29T03:03:26] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.00, R=0.00
2026-04-29T03:03:27] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:27] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:27] Watch HR:  81 BPM | Prototype HR:  86 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:27] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=5.91
2026-04-29T03:03:27] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.00, R=0.00
2026-04-29T03:03:27] Watch HR:  81 BPM | Prototype HR:  86 BPM | D=0.00, F=0.75, R=5.91
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=1.51, R=10.65
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.00, R=0.00
2026-04-29T03:03:28] Watch HR:  81 BPM | Prototype HR:  89 BPM | D=0.00, F=0.75, R=4.73
2026-04-29T03:03:29] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:29] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.00, R=0.00
2026-04-29T03:03:29] Watch HR:  81 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:29] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=5.91
2026-04-29T03:03:29] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:29] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:30] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=4.73
2026-04-29T03:03:30] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:30] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:30] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:30] Watch HR:  80 BPM | Prototype HR:  88 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:30] Watch HR:  80 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:31] Watch HR:  80 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=5.91
2026-04-29T03:03:31] Watch HR:  80 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:31] Watch HR:  80 BPM | Prototype HR:  82 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:31] Watch HR:  79 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=4.73
2026-04-29T03:03:31] Watch HR:  79 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:31] Watch HR:  79 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=5.91
2026-04-29T03:03:31] Watch HR:  79 BPM | Prototype HR:  80 BPM | D=0.00, F=0.75, R=5.91
2026-04-29T03:03:32] Watch HR:  79 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:32] Watch HR:  79 BPM | Prototype HR:  79 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:32] Watch HR:  78 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=7.10
2026-04-29T03:03:32] Watch HR:  78 BPM | Prototype HR:  95 BPM | D=0.00, F=0.75, R=4.73
2026-04-29T03:03:32] Watch HR:  78 BPM | Prototype HR:  87 BPM | D=0.00, F=0.75, R=3.55
2026-04-29T03:03:32] Watch HR:  78 BPM | Prototype HR:  95 BPM | D=0.00, F=0.75, R=4.73
2026-04-29T03:03:33] Watch HR:  78 BPM | Prototype HR:  95 BPM | D=0.00, F=0.75, R=3.55
```

Example:
1.2 Hz → 72 BPM

Dashboard with same timestamp



Only one channel activated right now in the image above

Next Steps

- Improve illumination/contact setup signal to find HR and remove noise.
- Test channel combinations
- Try longer windows for FFT-based HR estimation
- Decide best channel/settings combination

Code:
https://github.com/mharisirfan/heart-rate-logger

Thank you!